

Question (3): Best Rank-1 Approximation of A^2

We are given:

1. $A = PDP^T$, where:

- $P^{-1} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \frac{1}{2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{m} \end{bmatrix},$

- $D = \text{diag}(1, l^2, \dots, l^m)$ is a diagonal matrix with diagonal entries $1, l^2, \dots, l^m$,

- $|l| \neq 0$, so none of the diagonal entries of D are zero.

Step 1: Compute A^2

1. Using $A = PDP^T$, compute:

$$A^2 = A \cdot A = (PDP^T)(PDP^T) = PD(P^T P)DP^T.$$

2. Since $P^T P = I$ (identity matrix), this simplifies to:

$$A^2 = PD^2P^T.$$

Step 2: Compute D^2

1. Squaring a diagonal matrix means squaring each of its diagonal entries:

$$D^2 = \text{diag}(1^2, (l^2)^2, \dots, (l^m)^2).$$

2. So:

$$D^2 = \text{diag}(1, l^4, l^6, \dots, l^{2m}).$$

Step 3: Best Rank-1 Approximation

1. **Rank-1 Approximation:** A rank-1 matrix has the form:

$$uv^T,$$

where u and v are vectors.

2. The best rank-1 approximation of a matrix is based on its **largest eigenvalue** (in D^2) and the corresponding eigenvector.

3. **Largest Eigenvalue:**

- The eigenvalues of A^2 are the diagonal entries of D^2 :

$$1, l^4, l^6, \dots, l^{2m}.$$

- The largest eigenvalue is:

$$l^{2m}, \quad (\text{assuming } |l| > 1).$$

4. **Associated Eigenvector:**

- The eigenvector corresponding to the largest eigenvalue l^{2m} is the last column of P , denoted p_m .

5. **Rank-1 Approximation:**

- The best rank-1 approximation of A^2 is:

$$A^2 \approx l^{2m} p_m p_m^T.$$

Question (ii): Find a 2×2 Matrix A

We are given:

1. Vectors:

$$v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

2. Condition:

$$Av_i = -v_i, \quad i = 1, 2.$$

Step 1: Represent the Matrix A

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

When A acts on a vector v , it produces:

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}.$$

Step 2: Use the Condition for v_1

We are told $Av_1 = -v_1$. Substituting $v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$:

$$A \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} a \cdot \frac{1}{\sqrt{2}} + b \cdot \frac{1}{\sqrt{2}} \\ c \cdot \frac{1}{\sqrt{2}} + d \cdot \frac{1}{\sqrt{2}} \end{bmatrix} = - \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

This gives two equations:

$$a + b = -1, \tag{1}$$

$$c + d = -1. \tag{2}$$

Step 3: Use the Condition for v_2

We are told $Av_2 = -v_2$. Substituting $v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$:

$$A \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} a \cdot \frac{1}{\sqrt{2}} + b \cdot (-\frac{1}{\sqrt{2}}) \\ c \cdot \frac{1}{\sqrt{2}} + d \cdot (-\frac{1}{\sqrt{2}}) \end{bmatrix} = -\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

This gives two more equations:

$$a - b = -1, \tag{3}$$

$$c - d = 1. \tag{4}$$

Step 4: Solve the System of Equations

1. From (1) and (3):

- $a + b = -1,$
- $a - b = -1.$

Adding the equations:

$$2a = -2 \Rightarrow a = -1.$$

Subtracting the equations:

$$2b = 0 \Rightarrow b = 0.$$

2. From (2) and (4):

- $c + d = -1,$
- $c - d = 1.$

Adding the equations:

$$2c = 0 \implies c = 0.$$

Subtracting the equations:

$$2d = -2 \implies d = -1.$$

Step 5: Write the Matrix A

The matrix is:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Final Answers

1. Question (3): The best rank-1 approximation of A^2 is:

$$A^2 \approx l^{2m} p_m p_m^T.$$

2. Question (ii): The 2×2 matrix A is:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$